

US Stock Quant Toolkit: 美股量化策略回测研究项目

项目摘要

这是一个用于学习和研究的 Python 美股量化回测项目，可以下载真实市场数据，测试多种策略，并比较收益和风险。

数据来源

使用 yfinance 下载 SPY 或其他美股 ETF/股票的历史价格数据。

数据时间范围

Full period: 2018-01-02 到 2026-05-27。

Train period: 2018-01-01 到 2021-12-31。

Test period: 2022-01-01 到 2026-05-27。

策略设计

Buy and Hold、20/60 Moving Average、50/200 Moving Average、RSI Mean Reversion、6-Month Momentum。

回测方法

使用日收益率进行回测。所有交易信号使用 shift(1)，避免未来函数。交易成本 transaction_cost = 0.001。样本分为训练期和测试期，并重点分析测试期结果。

指标解释

指标包括 Cumulative Return、Annualized Return、Annualized Volatility、Sharpe Ratio、Maximum Drawdown、Calmar Ratio、Win Rate、Number of Trades 和 Turnover。

回测结果：全样本

Strategy	Cumulative Return	Annualized Return	Sharpe Ratio	Maximum Drawdown	Calmar Ratio
20/60 Moving Average	87.75%	7.81%	0.64	-27.12%	28.80%
50/200 Moving Average	103.14%	8.83%	0.57	-33.72%	26.18%
6-Month Momentum	72.28%	6.71%	0.54	-23.96%	28.00%
Buy and Hold	216.92%	14.76%	0.77	-33.72%	43.78%
RSI Mean Reversion	65.74%	6.22%	0.40	-28.34%	21.94%

回测结果：训练期和测试期

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RSI Mean Reversion	65.74%	6.22%	0.40	-28.34%	21.94%
20/60 Moving Average	65.43%	13.41%	1.04	-12.44%	107.82%
50/200 Moving Average	34.90%	7.77%	0.44	-33.72%	23.05%
6-Month Momentum	31.33%	7.05%	0.53	-23.96%	29.43%
Buy and Hold	89.21%	17.28%	0.83	-33.72%	51.26%
RSI Mean Reversion	32.31%	7.25%	0.43	-28.34%	25.58%
20/60 Moving Average	13.49%	2.93%	0.26	-26.42%	11.10%
50/200 Moving Average	50.59%	9.81%	0.72	-18.76%	52.28%
6-Month Momentum	31.18%	6.40%	0.56	-22.84%	28.01%
Buy and Hold	67.49%	12.51%	0.71	-24.50%	51.05%
RSI Mean Reversion	25.27%	5.28%	0.37	-19.29%	27.38%

主要结论

累计收益较高：Buy and Hold (216.92%)。

回撤较低：6-Month Momentum (-23.96%)。

Sharpe Ratio 较高：Buy and Hold (0.77)。

测试期 Sharpe Ratio 较高：50/200 Moving Average (0.72)。

交易成本会降低高换手策略的表现。

项目局限性

只使用历史数据，没有预测未来收益。没有考虑滑点、税费和流动性限制。单一资产测试不能代表所有市场环境。回测结果不等于未来表现。

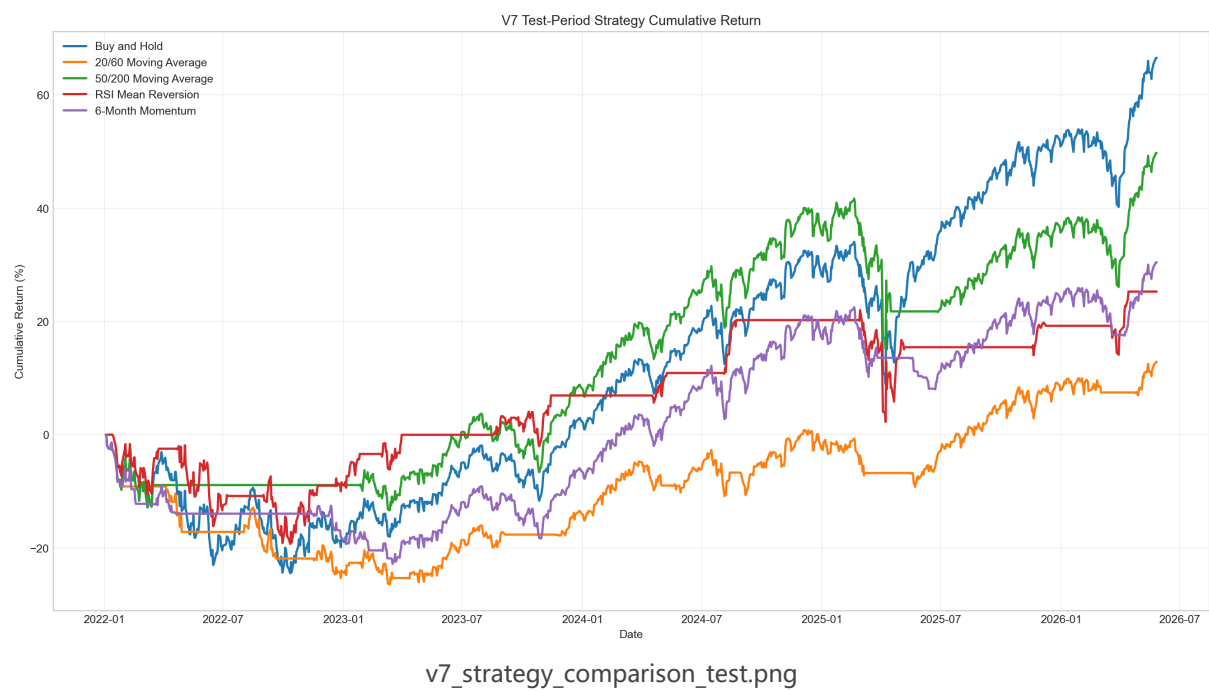
后续改进方向

可以加入多资产组合、风险平价组合、参数优化、Walk-forward validation、Monte Carlo simulation、宏观因子或机器学习模型。

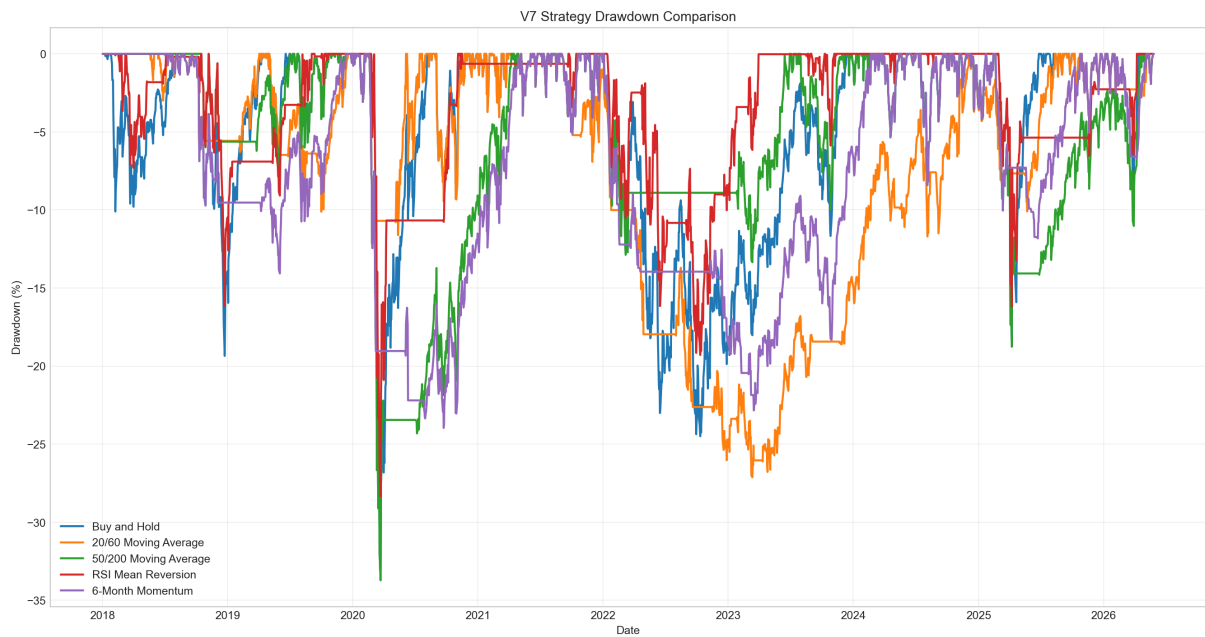
Charts



Full-sample cumulative return comparison.

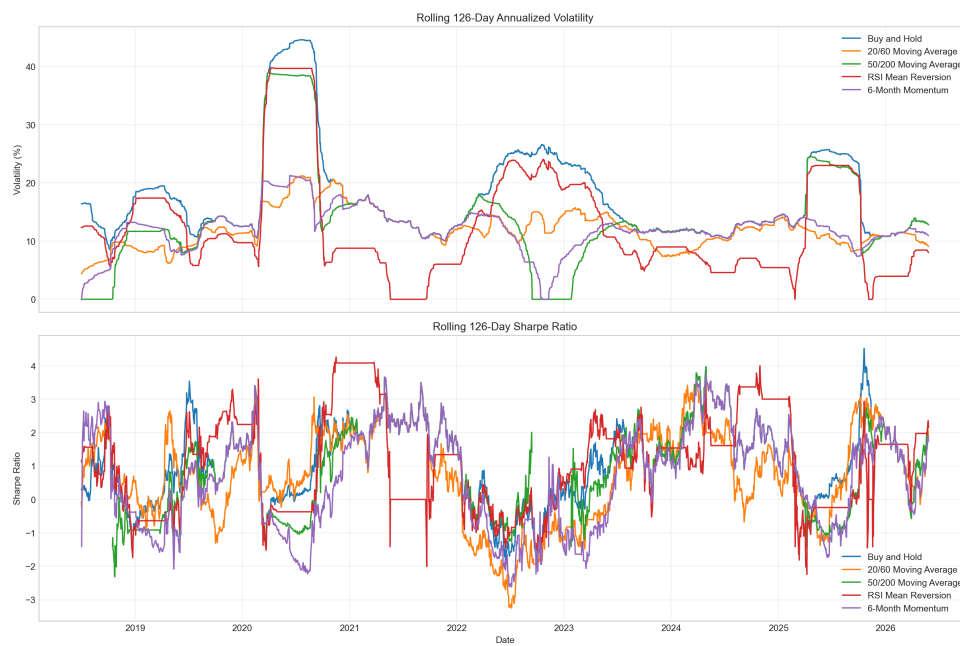


Testing-period cumulative return comparison.



v7_drawdown_comparison.png

Drawdown comparison across strategies.



v7_rolling_risk.png

Rolling volatility and rolling Sharpe Ratio.

Disclaimer

This project is for educational and research purposes only and does not constitute investment advice.